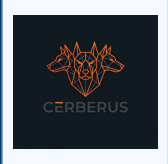


Cross-Asset Risk Divergence Engine



INTERNAL

Cross-asset structural divergence and re-alignment engine measuring risk-on vs risk-off spread.

Author: Cerberus Research

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CLASSIFICATION
internal

AUTHOR
Cerberus Research

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20260803_235040

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bccca1c0

DURATION
2.4s

DATASETS
market_ohlcv

PARAMETER · PERIOD
10y

PARAMETER · LOOKBACK
60

PARAMETER · DIVERGENCE_Z
2

PARAMETER · HORIZON
5

PARAMETER · SYMBOLS
['SPY', 'BTC-USD', 'TLT', 'DX-Y.NYB', 'GLD', 'LQD', 'HYG']

KEY FINDINGS

- Divergence events: 135 | Realignment probability: 50.4%
- Current divergence z: -0.60 | Spread (bp): -89.53

CONTENTS

- 01 Executive Summary
- 02 Results & Interpretation
- 03 Key Findings
- 04 Data Sources
- 05 Objectives
- 06 Methodology
- 07 Charts
- 08 Statistical Results
- 09 Limitations

Generated by the Cerberus studies framework · schema 2

Data Sources

- **market_ohlcv**: SPY, BTC-USD, TLT, DXY, GLD, LQD, HYG OHLCV via yfinance

Key Findings

- Divergence events: 135 | Realignment probability: 50.4%
- Current divergence z: -0.60 | Spread (bp): -89.53

Executive Summary

NOTE

Cross-asset structural divergence and re-alignment engine. Detects when risk-on and risk-off asset classes diverge and measures realignment probability.

Results & Interpretation

NOTE

Cross-asset structural divergence and re-alignment engine measuring risk-on vs risk-off spread.

Main takeaways:

- Divergence events: 135 | Realignment probability: 50.4%
- Current divergence z: -0.60 | Spread (bp): -89.53

Objectives

NOTE

Evaluate whether the studied signal produces a measurable, repeatable edge.

Quantify the reliability of the observed patterns across the sample.

Methodology

NOTE

Risk-on score computed as mean return of SPY and BTC-USD. Risk-off score as mean of inverted bonds, DXY, gold, and credit returns. Z-score of the spread identifies divergence events above threshold.

Charts

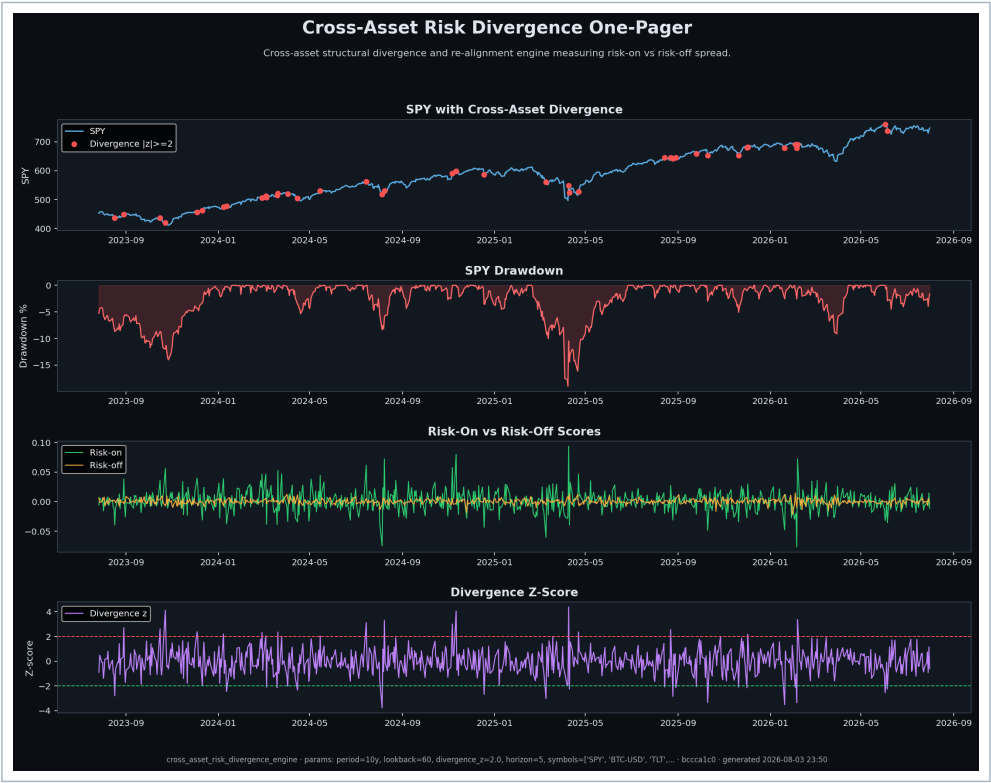


Figure 1. Cross-Asset Risk Divergence One-Pager
Cross-Asset Risk Divergence One-Pager

Statistical Results

time	z	spread_bp	reversion_score...	realigned
2016-11-03	-4.48	-406.06	246	True
2016-11-16	2.38	247.66	-34	False
2016-12-21	2.13	218.79	-736	False
2016-12-23	3.08	342.55	-624	False
2017-01-03	3.46	441.11	546	True
2017-01-04	3.96	582.58	1,820	True
2017-01-05	-3.96	-573.57	-1,090	False
2017-01-06	-3.44	-565.53	-413	False
2017-01-11	-3.62	-691.41	656	True
2017-01-17	2.31	513.92	99	True
2017-03-27	2.20	586.32	-478	False
2017-05-08	2.26	513.73	-97	False
2017-06-26	-2.70	-449.09	129	True
2017-07-20	4.49	1,219.10	306	True
2017-08-07	2.83	852.61	-1,257	False

time	z	spread_bp	reversion_score...	realigned
2017-08-14	2.94	966.12	422	True
2017-09-14	-2.95	-922.28	834	True
2017-09-15	2.19	769.11	-67	False
2017-11-27	2.78	926.17	-981	False
2017-12-06	2.83	1,009.79	-937	False
2017-12-07	3.31	1,281.42	298	True
2017-12-22	-2.27	-634.65	473	True
2018-01-08	-2.11	-650.74	-1,198	False
2018-01-16	-2.50	-883.13	-189	False
2018-02-05	-2.95	-1,262.06	1,267	True
2018-04-12	2.27	700.85	-238	False
2018-06-11	-2.27	-476.45	-104	False
2018-06-22	-2.11	-464.83	131	True
2018-07-16	2.08	392.37	-664	False
2018-07-17	2.34	458.26	-710	False
2018-07-24	2.58	504.74	372	True
2018-09-05	-2.08	-409.03	-350	False
2018-10-11	-2.12	-370.73	203	True
2018-11-14	-3.61	-535.52	-1,137	False
2018-11-19	-4.07	-705.27	-1,165	False
2018-11-20	-2.84	-544.51	65	True
2018-11-26	-2.70	-578.15	412	True
2018-11-28	3.47	686.71	944	True
2018-12-06	-2.23	-539.09	-369	False
2019-04-02	5.09	866.17	-332	False
2019-05-13	4.70	995.83	-222	False
2019-05-17	-2.09	-356.01	431	True
2019-06-27	-3.15	-651.38	131	True
2019-06-28	2.02	572.77	-139	False
2019-07-01	-2.89	-672.92	918	True
2019-07-16	-2.38	-670.31	257	True
2019-09-24	-2.14	-577.66	-145	False

Research only. Not investment advice. Past performance is not indicative of future results.

time	z	spread_bp	reversion_score...	realigned
2019-10-25	3.85	771.81	-469	False
2019-12-18	2.58	451.80	20	True
2020-01-14	2.20	429.74	0	True
2020-02-26	-2.16	-331.56	150	True
2020-03-09	-4.91	-1,017.29	-2,447	False
2020-03-12	-6.26	-2,365.82	764	True
2020-03-13	2.85	1,119.10	402	True
2020-03-16	-2.03	-869.02	699	True
2020-03-19	2.10	872.28	-1,329	False
2020-06-11	-2.17	-550.61	234	True
2020-07-27	3.50	752.87	-183	False
2020-09-03	-3.69	-659.21	-155	False
2020-10-21	2.13	352.07	-18	False
2020-11-05	3.14	588.91	-285	False
2020-11-27	-2.10	-355.51	454	True
2020-11-30	3.58	733.68	173	True
2020-12-16	2.05	473.54	-395	False
2020-12-28	3.11	759.94	-1,059	False
2021-01-11	-3.28	-696.91	165	True
2021-01-21	-2.75	-676.11	427	True
2021-02-08	3.16	1,091.18	-319	False
2021-04-19	-2.06	-492.87	-79	False
2021-05-12	-3.30	-803.15	-1,291	False
2021-05-17	-2.65	-683.03	-418	False
2021-05-19	-2.73	-725.84	507	True
2021-05-20	2.25	601.38	216	True
2021-06-09	2.23	605.81	-201	False
2021-09-20	-2.47	-531.49	-24	False
2021-10-01	2.36	584.49	-546	False
2021-11-08	2.17	518.89	515	True
2022-01-21	-2.65	-560.28	211	True
2022-02-04	3.03	540.21	85	True

Research only. Not investment advice. Past performance is not indicative of future results.

time	z	spread_bp	reversion_score...	realigned
2022-02-17	-2.30	-512.58	-152	False
2022-02-28	2.63	547.83	1,047	True
2022-03-09	2.75	634.60	8	True
2022-05-05	-2.43	-696.65	-1,223	False
2022-05-09	-2.88	-889.72	30	True
2022-06-13	-4.21	-1,416.30	-365	False
2022-08-19	-2.06	-637.85	-334	False
2022-09-09	2.54	622.59	634	True
2022-09-13	-2.86	-694.98	-482	False
2022-11-09	-3.07	-827.33	865	True
2022-11-10	3.54	968.62	143	True
2023-02-15	2.36	461.24	263	True
2023-03-09	-2.38	-416.79	1,217	True
2023-03-13	4.35	959.99	-767	False
2023-06-20	2.48	400.26	-418	False
2023-08-17	-2.81	-407.02	-65	False
2023-08-29	2.71	416.53	510	True
2023-10-16	2.61	325.32	-455	False
2023-10-23	4.12	637.19	-103	False
2023-12-04	2.37	422.81	-96	False
2023-12-11	-2.00	-291.98	422	True
2024-01-08	2.19	457.46	498	True
2024-01-12	-2.46	-402.25	-389	False
2024-02-28	2.32	475.58	-320	False
2024-03-04	2.05	411.86	-304	False
2024-03-05	-2.10	-346.42	656	True
2024-03-19	-2.15	-377.84	688	True
2024-03-20	2.36	514.55	-200	False
2024-04-02	-2.02	-386.70	242	True
2024-04-15	-2.35	-460.29	304	True
2024-05-15	2.03	484.90	-198	False
2024-07-15	3.12	562.76	-157	False

Research only. Not investment advice. Past performance is not indicative of future results.

time	z	spread_bp	reversion_score...	realigned
2024-08-05	-3.79	-698.77	525	True
2024-08-08	3.31	669.10	74	True
2024-11-06	3.01	518.06	-1,053	False
2024-11-11	4.06	827.68	58	True
2024-12-18	-2.69	-493.92	-159	False
2025-03-10	-3.03	-551.63	304	True
2025-04-09	4.38	903.70	230	True
2025-04-10	-2.27	-520.75	334	True
2025-04-22	2.36	507.79	-495	False
2025-08-14	-2.08	-223.37	-366	False
2025-08-22	2.56	309.16	447	True
2025-08-25	-2.85	-336.65	-139	False
2025-08-29	-2.14	-263.43	206	True
2025-09-25	-2.07	-244.15	576	True
2025-10-10	-3.35	-447.81	-299	False
2025-11-20	-2.10	-324.75	470	True
2025-12-01	-2.16	-348.95	287	True
2025-12-02	2.17	320.60	-59	False
2026-01-20	-3.52	-603.01	77	True
2026-02-03	-2.11	-383.82	-322	False
2026-02-05	-3.36	-668.28	372	True
2026-02-06	3.38	653.26	116	True
2026-06-02	-2.56	-316.49	-443	False
2026-06-05	-2.25	-308.38	354	True

Table 1. Divergence Events

Limitations

NOTE

Results are research-grade, not investment advice. The sample is limited to available data, and measured edges may not persist out-of-sample.

Appendix Tables

events

time	z	spread_bp	reversion_score...	realigned
2016-11-03	-4.48	-406.06	246	True
2016-11-16	2.38	247.66	-34	False
2016-12-21	2.13	218.79	-736	False
2016-12-23	3.08	342.55	-624	False
2017-01-03	3.46	441.11	546	True
2017-01-04	3.96	582.58	1,820	True
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2017-01-11	-3.62	-691.41	656	True
2017-01-17	2.31	513.92	99	True
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2017-05-08	2.26	513.73	-97	False
2017-06-26	-2.70	-449.09	129	True
2017-07-20	4.49	1,219.10	306	True
2017-08-07	2.83	852.61	-1,257	False

Statistics Summary

Key	Value
study	cross_asset_risk_divergence_engine
category	macro
status	success
parameters	{'period': '10y', 'lookback': 60, 'divergence_z': 2.0, 'horizon': 5, 'symbols': ['SPY', 'BTC-USD', 'TLT', 'DX-Y.NYB', 'GLD', 'LQD', 'HYG']}
datasets	['market_ohlcv']
rows	0
git_commit	bccca1c05d8a318e0779c4a3cf1e55db4dc14817
executed_at	2026-08-03T22:50:40.255448+00:00

Glossary

Term	Definition
n / count	Number of observations or events in the sample.
edge	Average excess return versus the unconditional baseline.
hit-rate	Share of events with a positive forward return.
p-value	Probability of observing the result under the null of no edge.
*** / ** / *	Significance at 99.9% / 99% / 95% confidence.
α (alpha)	Intercept or active return versus benchmark.
β (beta)	Sensitivity to a reference factor or market.
σ (sigma)	Volatility / standard deviation of returns.

Document Control

Field	Value
Report schema	2
Framework version	0.1.0
Classification	internal
Generated	2026-08-03 22:50 UTC
Author	Cerberus Research
Study slug	cross_asset_risk_divergence_engine
Run ID	20260803_235040
Git commit	bccca1c0
Datasets	market_ohlcv
Parameters hash	8f5d98f25a17
Parameter · period	10y
Parameter · lookback	60
Parameter · divergence_z	2
Parameter · horizon	5
Parameter · symbols	['SPY', 'BTC-USD', 'TLT', 'DX-Y.NYB', 'GLD', 'LQD', 'HYG']
Data fingerprint	5a84d41f9669
Duration	2.4s